# SPEC CPU2017 Spill Metrics - Schema-v4 Quick Results

Generated 2026-09-16 10:04 +03

Three short SPEC CPU2017 workloads completed successfully with the corrected gem5 spill detector.

## Experimental scope

Simulator: gem5 RISC-V TimingSimpleCPU, private L1 caches, L2 cache, 4 GiB memory.

Detector: heuristic_stack_store_reload_match; metric schema 4.

gem5 commit: 62d20b39c49f.

Selection: the three fastest available small-input workloads based on earlier host-time observations. Existing valid GCC output was reused; missing Perlbench and Imagick outputs were run for this report.

| Benchmark | Input | Duration | State | Classification |
|---|---|---|---|---|
| GCC | t1.c, -O3 | 00:00:09 | COMPLETED | heuristic_stack_store_reload_match |
| Perlbench | test.pl | 00:00:09 | COMPLETED | heuristic_stack_store_reload_match |
| Imagick | test_input.tga, short filter chain | 00:00:54 | COMPLETED | heuristic_stack_store_reload_match |

LIMITATION: These are small functional workloads, not official SPEC reportable performance runs. The heuristic identifies qualifying stack store-to-later-load reuse; it is not compiler ground truth. Conclusions apply only to these executions.

## Dynamic instruction results

| Benchmark | ROI inst | Loads | Stores | Events | Spill loads | Conf stores | Spill mem | Share |
|---|---|---|---|---|---|---|---|---|
| GCC | 16,026,850 | 2,957,472 | 1,955,133 | 1,020,580 | 1,041,215 | 1,020,580 | 2,061,795 | 12.8646% |
| Perlbench | 14,690,833 | 3,491,472 | 2,035,263 | 1,049,550 | 1,074,110 | 1,049,550 | 2,123,660 | 14.4557% |
| Imagick | 117,737,096 | 20,614,195 | 8,988,558 | 2,444,450 | 2,466,889 | 2,444,450 | 4,911,339 | 4.1714% |
| Weighted total | 148,454,779 | 27,063,139 | 12,978,954 | 4,514,580 | 4,582,214 | 4,514,580 | 9,096,794 | 6.1277% |

spill memory instructions = spill load instructions + confirmed spill store instructions
spill instruction share = spill memory instructions / ROI instructions

The weighted spill share is recomputed from summed populations; it is not an arithmetic mean.

## Candidate and confirmed classification

| Benchmark | Candidate stores | Confirmed stores | Unconfirmed | Repeated loads |
|---|---|---|---|---|
| GCC | 1,103,563 | 1,020,580 | 82,983 | 20,635 |
| Perlbench | 1,118,231 | 1,049,550 | 68,681 | 24,560 |
| Imagick | 2,453,995 | 2,444,450 | 9,545 | 22,439 |
| Total | 4,675,789 | 4,514,580 | 161,209 | 67,634 |

A candidate becomes a confirmed spill store only when its first qualifying later load is observed. Further matching loads of the same store generation are repeated spill loads.

## Confirmed spill-load cache behavior

| Benchmark | L1 hit | L1 miss | L1 hit % | L2 hit | L2 miss | L2 local % | L2/L1 % | DRAM/L1 % |
|---|---|---|---|---|---|---|---|---|
| GCC | 1,041,089 | 126 | 99.9879% | 124 | 2 | 98.4127% | 0.0119% | 0.0002% |
| Perlbench | 1,074,026 | 84 | 99.9922% | 84 | 0 | 100.0000% | 0.0078% | 0.0000% |
| Imagick | 2,463,761 | 3,128 | 99.8732% | 3,102 | 26 | 99.1688% | 0.1257% | 0.0011% |
| Weighted total | 4,578,876 | 3,338 | 99.9272% | 3,310 | 28 | 99.1612% | 0.0722% | 0.0006% |

## Confirmed spill-store cache behavior

| Benchmark | L1 hit | L1 miss | L1 hit % | L2 hit | L2 miss | L2 local % | L2/L1 % | DRAM/L1 % |
|---|---|---|---|---|---|---|---|---|
| GCC | 1,020,064 | 516 | 99.9494% | 447 | 69 | 86.6279% | 0.0438% | 0.0068% |
| Perlbench | 1,048,011 | 1,539 | 99.8534% | 1,491 | 48 | 96.8811% | 0.1421% | 0.0046% |
| Imagick | 2,442,175 | 2,275 | 99.9069% | 1,915 | 360 | 84.1758% | 0.0783% | 0.0147% |
| Weighted total | 4,510,250 | 4,330 | 99.9041% | 3,853 | 477 | 88.9838% | 0.0853% | 0.0106% |

## Candidate-store cache behavior

| Benchmark | L1 hit | L1 miss | L1 hit % | L2 hit | L2 miss | L2 local % | L2/L1 % | DRAM/L1 % |
|---|---|---|---|---|---|---|---|---|
| GCC | 1,102,960 | 603 | 99.9454% | 508 | 95 | 84.2454% | 0.0460% | 0.0086% |
| Perlbench | 1,116,309 | 1,922 | 99.8281% | 1,864 | 58 | 96.9823% | 0.1667% | 0.0052% |
| Imagick | 2,450,836 | 3,159 | 99.8713% | 2,634 | 525 | 83.3808% | 0.1073% | 0.0214% |
| Weighted total | 4,670,105 | 5,684 | 99.8784% | 5,006 | 678 | 88.0718% | 0.1071% | 0.0145% |

## Denominators and interpretation

- L1 hit rate = L1 hits / (L1 hits + L1 misses)
- L2 local hit rate = L2 hits / (L2 hits + L2 misses)
- L2 rescue / L1 = L2 hits / L1 accesses
- DRAM / L1 = L2 misses / L1 accesses

All weighted cache percentages are recomputed from summed hit/miss populations. Split memory instructions can produce multiple cache accesses, so cache rates use access counts rather than instruction counts.

## Reproduction

./spec_spill_runner.sh status test standard gcc perlbench imagick
./spec_spill_runner.sh results test standard gcc perlbench imagick

Source m5out directories:
• GCC: /cta/users/bkaya/riscv-spec-register-spilling/gem5/benchmarks/gcc/m5out_spill_v4/standard/test/gcc/20260908-170008
• Perlbench: /cta/users/bkaya/riscv-spec-register-spilling/gem5/benchmarks/perlbench/m5out_spill_v4/standard/test/perlbench_test/20260916-095531
• Imagick: /cta/users/bkaya/riscv-spec-register-spilling/gem5/benchmarks/imagick/m5out_spill_v4/standard/test/imagick/20260916-095531